

Ankit Kumar

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Summary

Robotics M.S. graduate specializing in perception, SLAM, and autonomous navigation for drones and mobile robots. Experienced in stereo vision, LiDAR/RGB-D mapping, deep learning, and real-time robotics with Python, C++, and ROS 2.

Education

University of Minnesota

M.S. in Robotics, GPA: 3.78/4.0

Twin Cities, MN

Sep 2024 - Dec 2026

Coursework: Intelligent Robotic Systems, Robot Vision, Machine Learning, Deep Learning, Computer Vision

Jadavpur University

B.E. in Mechanical Engineering, GPA: 8.55/10

Kolkata, India

Aug 2018 - May 2022

Technical Skills

Programming: Python, C++, MATLAB, Bash

Perception & Vision: Stereo Vision, Camera Calibration, Triangulation, Object Detection & Tracking, Semantic Segmentation

SLAM & Navigation: RGB-D SLAM, RTAB-Map, LiDAR Localization, AMCL, Occupancy-Grid Mapping, Nav2, Path Planning

Robotics & Simulation: ROS/ROS 2, Gazebo (Classic/Harmonic), NVIDIA Isaac Sim, MAVROS, PX4, ArduPilot

Machine Learning: PyTorch, YOLO, U-Net, CNNs, TensorRT, KMeans Clustering, Synthetic Data Augmentation

Embedded & Tools: Jetson Orin Nano, OpenCV, COLMAP, Instant-NGP, Linux, Git, GStreamer

Experience

Flow Field Imaging Lab, University of Minnesota

Twin Cities, MN

Graduate Research Assistant

May 2025 - Present

Wildfire Ember Perception and Autonomous Tracking

- Developed an end-to-end real-time perception and control pipeline on a custom drone using synchronized stereo cameras and an onboard Jetson Orin Nano for wildfire ember tracking in real time.
- Designed and trained a multi-head U-Net with a custom multi-task loss, achieving 90-95% detection accuracy and improving small-object recall under motion blur and low-contrast conditions during system evaluation.
- Developed a calibrated stereo vision pipeline with triangulation and temporal association, achieving less than 10 cm 3D trajectory error for real-time tracking and motion estimation on the drone platform.
- Integrated perception with the drone control loop and optimized embedded deployment to approximately 10 FPS, improving robustness by 20-30% through synthetic data augmentation during testing.

Multi-UAV Autonomous Field Deployment

- Built a ROS 2/MAVROS four-drone formation system with PX4 and RTK-GPS for synchronized waypoint missions.
- Synchronized RGB-DIH imagery and telemetry across four UAVs to analyze smoke-dispersion dynamics.
- Integrated RTK-GPS and flight telemetry to support coordinated outdoor deployment and repeatable field testing.

Projects

Autonomous Mobile Robot Navigation & SLAM | ROS 2, RTAB-Map, Nav2, LiDAR

Jan 2026 - May 2026

- Developed a multi-sensor autonomous mobile-robot pipeline in NVIDIA Isaac Sim integrating RTAB-Map RGB-D SLAM, LiDAR localization, and Nav2 waypoint navigation using ROS 2.
- Implemented a custom four-wheel differential-drive controller and Dijkstra-based path planning, achieving approximately 0.82 m ATE RMSE and 0.19 m RPE RMSE using evo trajectory evaluation.
- Integrated occupancy-grid mapping and AMCL localization to support autonomous indoor navigation.

NeRF-Based 3D Reconstruction from Drone Imagery | ROS, OpenCV, PyTorch, Instant-NGP

Sep 2024 - Dec 2024

- Developed autonomous target-circling and viewpoint-selection logic for high-resolution multi-view image capture.
- Built a COLMAP and Instant-NGP reconstruction pipeline with image filtering, target cropping, and hash-grid encoding for high-fidelity 3D reconstruction from autonomous drone imagery.

VTOL Autonomous Flight Simulation | Gazebo Harmonic, ROS 2, ArduPilot

May 2025 - Aug 2025

- Built a Gazebo Harmonic-ArduPilot SITL framework with ROS 2 and MAVROS for autonomous VTOL flight testing.
- Ran QGroundControl waypoint missions to validate fixed-wing navigation, trajectory tracking, and closed-loop flight control.

Football Player Tracking & Analysis | YOLO, OpenCV, KMeans

Oct 2024 - Dec 2024

- Developed a YOLOv8 multi-class detection and tracking system for dynamic video scenes, using KMeans for team classification and optical flow for camera-motion analysis.
- Applied perspective transformation to convert image-space motion into real-world distance and velocity estimates.